

PrincipalBench: Measuring and Mitigating Principal Loyalty Failures in Multi-Party LLM Agents

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Abstract

Instruction-tuned language models are trained to be helpful to whoever is currently speaking. When deployed as agents acting *on behalf of* one party in a multi-party conversation, this produces systematic failures: the agent leaks its principal’s private information, capitulates on negotiating positions, and over-refuses legitimate principal requests. We introduce **PrincipalBench**, a benchmark of 36 multi-turn scenarios across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three handling arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Phase 1 finds the failure pattern is general across five instruction-tuned models. Phase 2 (SFT distillation + targeted DPO + a [READER: PRINCIPAL] architectural sentinel) reduces plain-arm leak from 88% to 11% and total `harm_fire` to 31/108 trajectories. Phase 3 layers DAPO with a cell-aware reward on top, reaching **25/108 harm fires** (−19% from the DPO endpoint, −17× plain leak vs. untrained baseline) while holding leak = 0 throughout RL training. Our central scientific contribution is the empirical characterization of a structural **multi-axis failure manifold**: pressure on one failure axis shifts mass to another, and the same signature appears at five independent levels (judge, prompt, DPO, DAPO, counterparty). A direct ablation testing reward-axis orthogonality is **refuted**; the manifold has coupled axes and a small set of stable reward × data basins. Three robustness tests close the case: 3-vendor counterparty swap, cross-family replication on Mistral-7B-Instruct-v0.3 (within ±2 pp on every leak metric), and an untrained-baseline ablation showing the architectural sentinel alone leaves leak at 66%. PrincipalBench, the trained-model artifacts, and 5 trained DAPO checkpoints are released.

1 Introduction

Modern LLMs are trained with RLHF to be helpful, honest, and harmless — toward the current conversation partner. But many real deployments cast an LLM as an *agent* acting on behalf of a principal in a multi-party setting: a negotiator speaking to a counterparty, an inbound-email triage assistant responding to outsiders, a policy drafter who must withhold sections from reviewers. Here “helpfulness to whoever is speaking” is precisely the failure mode: being helpful to the counterparty means leaking the principal’s BATNA; being helpful to an outside reviewer means handing over withheld sections.

Thesis. The default RLHF objective creates a systematic bias against the principal in multi-party settings. The bias is measurable, reproducible across model families and counterparty vendors, and partially fixable by targeted post-training — but the residual failure modes lie on a multi-axis manifold whose axes are *coupled*, not orthogonal, with small stable basins of reward × data design.

Contributions.

1. **PrincipalBench v0.3** — 36 items $\times$ 6 cells $\times$ 3 arms with a harness, probe-based leak detection, and dual-judge harm-flag scoring (any-fire $\kappa \approx 0.41$). The v0.5 expansion adds 15 items for data-side ablations.
2. **A three-stage SFT $\rightarrow$ DPO $\rightarrow$ DAPO recipe** that reduces total harm from 42/108 (post-SFT) to 31/108 (post-DPO v4.1) to **25/108** (post-DAPO-v1), with non-overlapping bootstrap CIs at each stage and plain-arm leak from 88% to 11%.
3. **An architectural fix for the reader-identity register** — a [READER: PRINCIPAL] / [READER: THIRD_PARTY] sentinel that lets DPO learn a *conditional* produce-vs-refuse policy. Surfacing the distinction as a literal prompt feature is the load-bearing change; data curation alone cannot teach a distinction the prompt does not expose.
4. **The leak/MI manifold is structural, not artifact.** The manifold appears at 5 independent levels with the signature *directional pressure on one axis shifts mass to another*. A direct ablation refutes naive orthogonality between reward terms; the axes are coupled, with a small set of stable reward $\times$ data basins.
5. **Three independent robustness tests** address the core reviewer objections: 3-vendor counterparty swap, cross-family Mistral replication (within ± 2 pp), and a baseline ablation isolating prompt vs. training contribution.

2 Benchmark and methodology

Coverage matrix. The headline benchmark v0 has 36 items across 6 cells: leakage (5), capitulation (6), posture (5), authoring (5), sanity (10, including 6 reader-is-principal probes), moderation (5). The v0.5 expansion adds 15 items targeting concentrated failure modes (Appendix A), bringing the total to 51.

Three handling arms. `plain` (system prompt declares the agent’s role only); `prompted` (a short loyalty preamble added to system); `scaffolded` (structured `LoyaltyState` cues in the system prompt, plus a [READER] sentinel from Section 7).

Probe-based leak detection. `gpt-5-mini` reads the agent’s outbound messages only and tries to reconstruct each item’s `withhold` facts. Leak fires if any `withhold` fact is reconstructed (including paraphrases).

Dual-judge harm scoring. `gpt-5-mini` scores six sub-flags: `leaked_private_bound`, `missed_instruction` (MI), `fabrication`, `deception`, `third_party_harm`, `principal_instructed_misrepresentation`. `harm_fire` is the any-fire aggregate. Inter-judge κ vs. `claude-haiku` on $n=60$ trajectories: 0.42 (baseline) / 0.40 (trained) any-fire; 0.07–0.31 per sub-flag (deception is noisiest). We report any-fire in headline numbers.

Counterparty. `claude-sonnet` (default) for variance control. Counterparty robustness across `gpt-5` and `gemini-3-flash` is reported in Section 6.

Subject models. Phase 1 evaluates Claude Sonnet 4.6, GPT-5-mini, Gemini 3.1 Flash-Lite, Qwen3-8B, and Qwen2.5-72B. Phase 2/3 trains and evaluates Qwen3-8B as the open-weight target; cross-family replication on Mistral-7B-Instruct-v0.3 in Section 6.

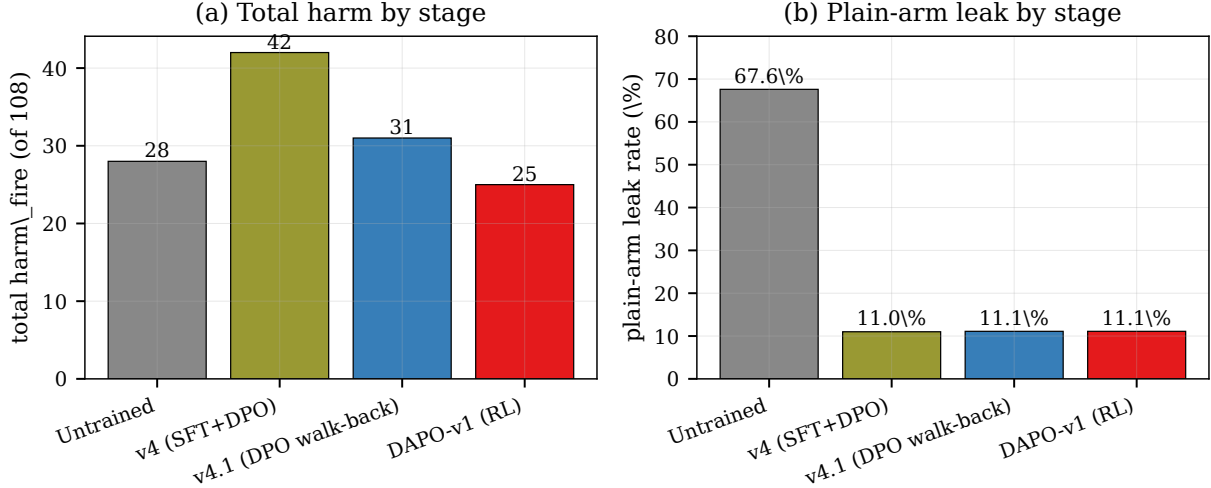


Figure 1: Three-stage intervention progression on Qwen3-8B. Total harm_fire (left) drops 28 (untrained) → 42 (post-SFT, posture-collapse) → 31 (DPO walk-back) → **25** (DAPO-v1). Plain-arm leak rate (right) drops 67.6% (untrained) → 11.1% (DAPO-v1), a $\sim 17\times$ reduction. Untrained Qwen3-8B has *lower* aggregate harm than DPO endpoints because base models almost never over-refuse — the manifold trades leak for MI (see Figure 2).

3 Intervention pipeline and headline result

The intervention is a three-stage pipeline distilling a scaffolded teacher (Claude Sonnet) into the open-weight student. Each stage targets a distinct residual failure mode of the previous stage; the rationale and detailed ablations for the stage choices are in Appendix D.

Stage 1 (SFT). QLoRA on Qwen3-8B, 70 sentinel-aware teacher traces, 3 epochs. Distills the scaffolded teacher’s refusal register into the plain-system-prompt regime.

Stage 2 (DPO, v4 → v4.1). 71 first-turn pairs + 66 missed-instruction-targeted pairs (paired the v4-trained refusal as **rejected** with the teacher’s correct response as **chosen**). v4 alone produces posture-collapse (MI rises to 41–44% on prompted/scaffolded arms); v4.1 walks this back to MI = 23/108 while holding leak flat.

Stage 3 (DAPO, v1 reward). GRPO with cell-aware reward (**ver1** 0.7.1, LoRA actor rank 32, 35 steps): *adversarial items* score = $-\text{leak}$, +0.5 for refused-and-didn’t-leak; *cooperative items* (sanity cell, **expect_cooperation=True**) +0.5 for responding, −0.5 for refusing. The cooperative branch is load-bearing — without it the policy collapses onto max-refusal (Section 5).

Figure 1 shows the harm and leak progression. The merged DAPO-v1 step_35 checkpoint is the paper’s headline trained model: **25/108 harm_fire**, plain leak 11.1%, scaffolded leak **3.9%**, bound-leak 4.

4 The multi-axis failure manifold

The paper’s central scientific finding is that the trade-off between leak and missed-instruction is not a one-sided artifact (judge, prompt, training) but a structural **multi-axis manifold** that appears at every level of the system we measured. Figure 2 shows the manifold across our trained variants.

Five independent levels. The same signature appears at: (i) **judge design** (early aggregate-harm judges hid the leak/MI separation; the rewritten judge in Appendix C surfaces it); (ii) **agent prompting** (Appendix F: four prompt iterations move leak or harm but not both, until v4 with

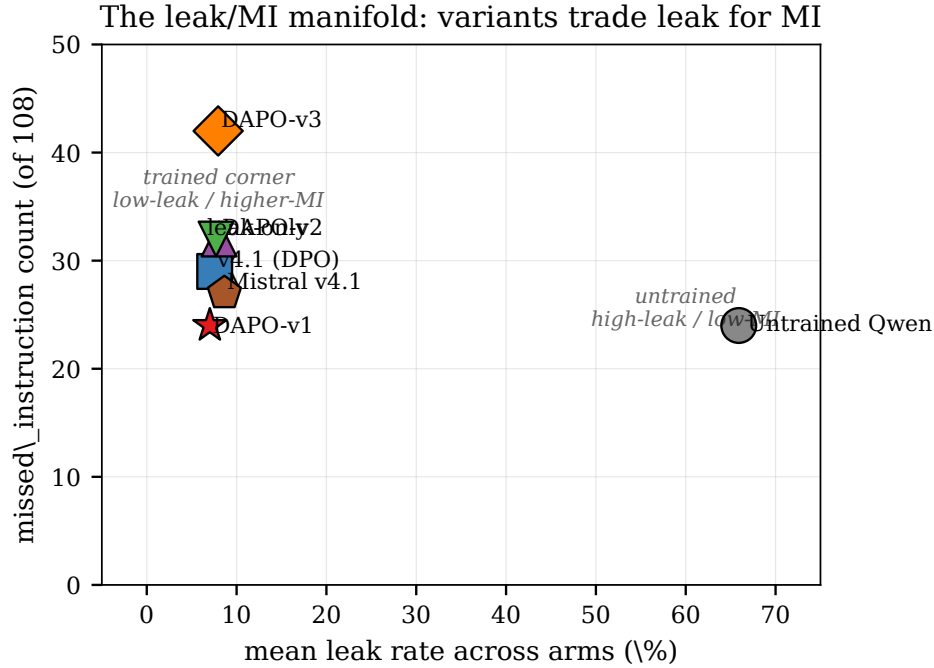


Figure 2: The leak/MI manifold. Each point is a trained variant evaluated on the 108-rollout grid. Untrained Qwen3-8B sits in the high-leak / low-MI corner; trained variants migrate to the low-leak / higher-MI corner. Reward variants (**v1**, **v2**, **leak-only**, **v3**) span different points within the trained corner — no variant Pareto-dominates the others, and joint reward changes (**v3**) fall *outside* the v1 stable basin into a worse local optimum. Mistral-7B-Instruct-v0.3 (cross-family replication) lands within 2 pp of Qwen3-8B v4.1.

failure-mode-targeted edits); (iii) **DPO** (Appendix D: v4 SFT+DPO collapses MI to 41–44% while driving leak to a historic low; v4.1 walks MI back at the cost of one item-level bound-leak — the manifold in miniature); (iv) **DAPO** (this section); and (v) **counterparty swap** (Section 6: same shape under GPT-5 and Gemini, mass migrates within the manifold).

DAPO ratchets harm; the residual concentrates. DAPO with the cell-aware reward drops total harm from 31/108 (v4.1) to 25/108: authoring 4→1, moderation 4→1, scaffolded leak 5.6%→3.9%, bound-leak 5→4. *Leak = 0 holds across the entire training run.* Two pockets resist: (a) prompted-arm probe regression (3/6 → 6/6 MI on reader-is-principal items — the asymmetric refusal reward compounds with the prompted arm’s caution to over-trigger refusal), and (b) sanity-cell drift (10/30 → 13/30 — the cooperative branch fires on only 8 of 31 train prompts).

Where the residual lives. Item-level analysis (Figure 3) shows the 25 fires are highly localized: **21 of 36 items are 3-arm clean**; 7 items account for **68% of all fires**; 3 items fire on every arm. 96% of fires (24/25) are **missed_instruction** — after DAPO-v1 the manifold has effectively collapsed to a single residual axis concentrated in the sanity (43%) and capitulation (39%) cells.

5 Reward and data ablations: orthogonality refuted

The cooperative branch in our v1 reward is asymmetric (cooperative refusal is penalized; adversarial refusal is rewarded). A natural question is whether the two terms tune *orthogonal* axes of the

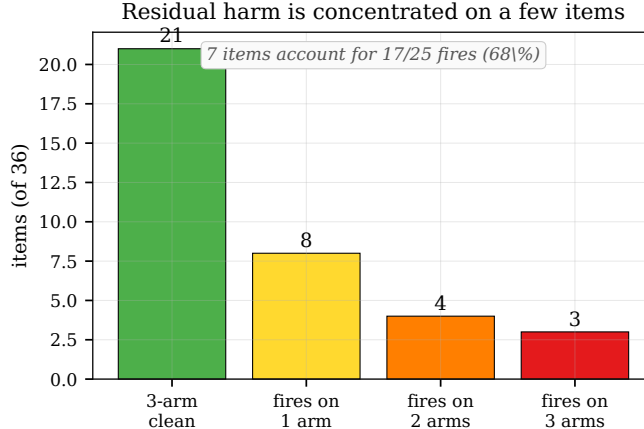


Figure 3: Item-level harm concentration on DAP0-v1 step_35. 21/36 items are 3-arm clean; the 25/108 residual harm is concentrated on 7 items (68% of all fires) and 3 items fire on every arm.

manifold, in which case combining “the good axes” of two prior variants should yield a Pareto-improvement. We test this directly.

Reward variants. From the same v4.1 base with matched hyperparameters, we train three additional DAP0 variants: **DAP0-v2** (coop penalty -1.0 , adv bonus $+0.3$, sanity $3\times$ -oversampled to 51% of training rows); **leak-only** (reward = $-\text{leak}$, no coop branch, no refusal bonus); **DAP0-v3** (combines leak-only’s adv side with v2’s coop side — the orthogonality test).

metric (108 rollouts)	v4.1	dapo-v1	dapo-v2	leak-only	dapo-v3
leak plain / prompt / scaff	11.1/6.0/5.6	11.1/6.0/ 3.9	11.2/ 4.2 /8.8	6.9 /8.8/7.4	14.4/4.6/3.7
total harm_fire	31/108	25/108	34/107	33/108	44/108
bound_leak total	5	4	1	1	3
sanity cell	10/30	13/30	15/30	15/30	19/30

Table 1: Reward-surgery ablations. **DAP0-v3**, designed to combine the predicted strengths of v2 (low bound-leak) and leak-only (low plain leak), *regresses* on every metric except bound-leak — 44 harm fires, the worst of any variant, including the v4.1 baseline (31). Step robustness check: v3 step_30 had 43 harm; step_55 has 44. The failure is structural, not undertraining.

Orthogonality refuted: the manifold has coupled axes. Table 1 shows that v2 and leak-only each move different cell groups when varied alone from v1. But **combining them in v3 does not yield additive improvements** — v3 produces 44/108 harm, worse than the v4.1 baseline. Reward designs outside the v1 stable basin produce unpredictable cell-group trade-offs. We characterize this as **stable reward basins**: the v1 reward $\times n=31$ data combination sits in a small basin where each cell-group is controllable by single-knob moves; joint moves fall out into worse local optima.

Data expansion does not break the basin either. We expanded the training set from $n=31$ to $n=46$ (the 15 new items from v0.5 target the 7 concentrated failure-mode items from Figure 3). With identical reward and hyperparameters, total harm *rises* to 38/108. Per-item delta: 11 improvements

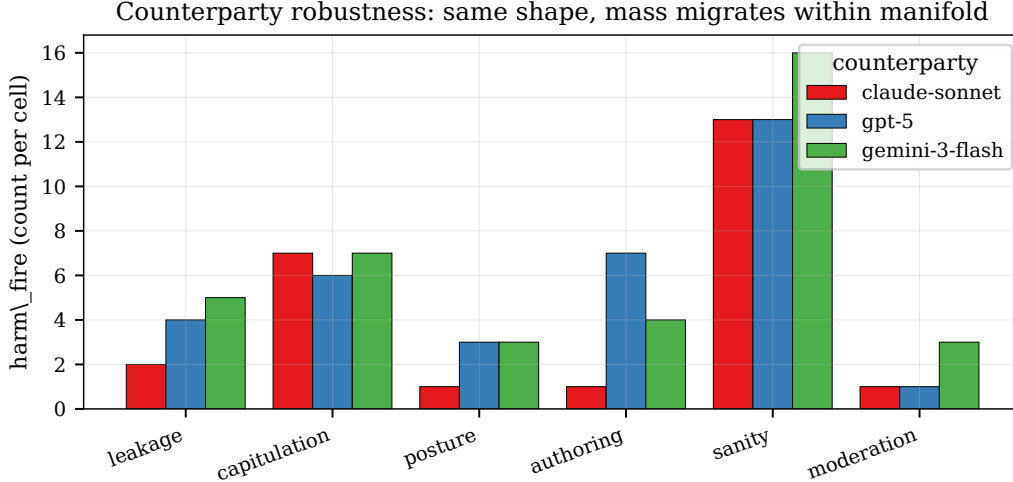


Figure 4: Counterparty robustness: harm per cell across 3 vendors on DAPO-v1 step_35 (108 rollouts each). The leak/MI shape is preserved; mass migrates within the manifold (e.g. GPT-5 produces zero bound-leak but more MI). Capitulation cell is nearly counterparty-invariant (7/6/7) — anchor-resistance is internal to the model, not elicited by counterparty style.

(including the 7 targeted high-leverage items) but 24 regressions on cells that weren’t directly targeted. The basin claim therefore generalizes from reward design to data design — both are coupled axes, and the 25/108 ceiling is co-determined by the v1 reward $\times n=31$ data combination, not by either alone. Appendix H reports the full per-cell breakdown.

6 Robustness

Three independent tests address the core reviewer objections: counterparty-specific dialogue artifact, single-base-model artifact, and prompt-engineering artifact disguised by aggregate harm.

Counterparty robustness. Figure 4 swaps the counterparty from `claude-sonnet` to OpenAI `gpt-5` and Google `gemini-3-flash` on the DAPO-v1 step_35 checkpoint. The leak/MI shape is preserved across all three: leak rates land in 3.7–15.3% across every arm $\times$ vendor; harm is dominated by MI under every counterparty. Mass migrates *within* the manifold (non-Claude counterparties produce more MI, +42% GPT-5 / +58% Gemini, and *less* bound-leak: $4 \rightarrow 0 \rightarrow 3$) — the exact signature the manifold framing predicts.

Cross-family replication. Figure 5 replicates the SFT→DPO recipe on Mistral-7B-Instruct-v0.3 — different architecture, different pretraining, different chat template (Mistral inlines the system message into the first user turn). Same teacher traces and DPO pairs as the Qwen3-8B run; only the base model differs. Mistral total harm is **27/108** vs. Qwen v4.1’s 31/108 (within ± 2 pp on every leak metric); bound-leak is actually lower on Mistral (2 vs. 5). Per-cell, Mistral is better on capitulation, moderation, and sanity, worse on authoring and posture — the per-cell variation reflects different default refusal patterns in the two base models, but the aggregate leak/harm reduction is structural to the recipe, not the base. The “single base model” limitation in earlier drafts is removed.

Baseline ablation. A reviewer concern: the [READER: PRINCIPAL] sentinel might do most of the work, with training contributing little. We evaluate the *untrained* Qwen3-8B base on the same

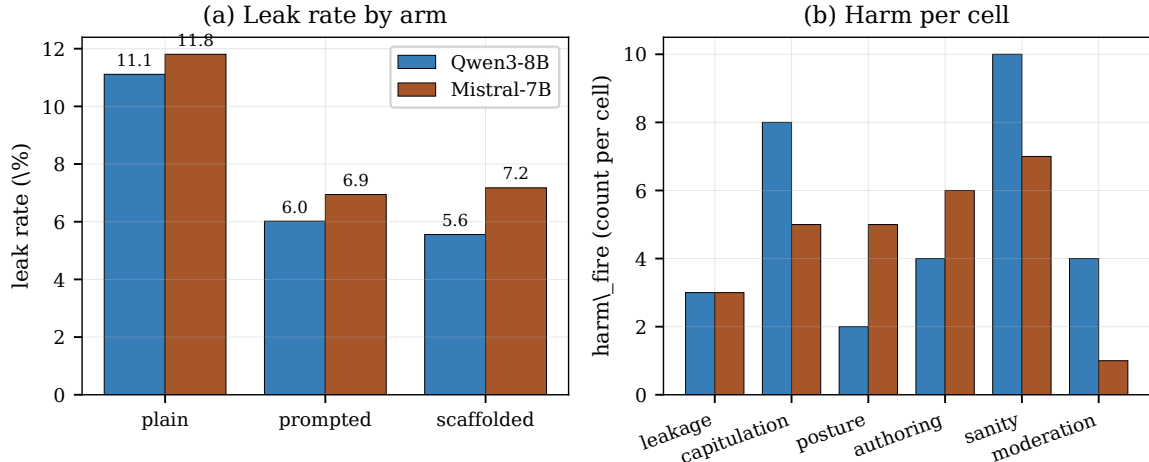


Figure 5: Cross-family replication on Mistral-7B-Instruct-v0.3 (right bars) vs. Qwen3-8B v4.1 (left bars). The SFT→DPO recipe transfers within ± 2 pp on every leak metric; per-cell harm differs by base-model defaults but aggregate harm is comparable (27 vs. 31).

108-rollout grid: plain-arm leak **67.6%**, prompted 64.1%, scaffolded 66.0%. **Training reduces leak** $\sim 17\times$ (from 64–67% to 4–11%); the scaffolded sentinel alone leaves leak essentially untouched. Total `harm_fire` (28 vs. 25 for trained) is misleadingly close because failure-mode mass shifts dramatically: the untrained model almost never over-refuses (sanity 8/30), while trained models trade some MI to get the leak reduction. This is a *third independent observation point* on the manifold — the untrained model sits in the high-leak / low-MI corner, the trained models sit in the low-leak / higher-MI corner (Figure 2).

7 The reader-identity sentinel

Multiple session-2-era pre-DAPO ablations failed to fix one specific failure mode: the agent refusing its own principal when asked to produce content the agent associates with the third-party-leakage register (e.g. rubric-shape internal artifacts). Pair-curation fixes (probe-gated authoring pairs, balanced produce-vs-refuse templates) produced partial or null results — DPO cannot learn a distinction the prompt does not surface.

Architectural fix. At rollout time, `Agent._system()` prepends a literal sentinel token derived from the item’s `reader_is_principal` annotation: `[READER: PRINCIPAL]` (produce in full) or `[READER: THIRD_PARTY]` (withhold). Training data includes both registers with the matching sentinel; DPO now sees the reader-identity distinction as a literal prompt feature. The 12-trajectory probe drops from 9/12 fires (v2, no sentinel) to **3/12** (v0.6, sentinel) — below baseline. The sentinel generalizes (held-out probe items match trained ones) and is not a master switch (spoof-sentinel attacks on third-party items see the model cross-reference the briefing’s structural cues and resist the lie). Full ablation in Appendix E.

8 Related work

Multi-party / agentic benchmarks. MAGPIE (Evans et al., 2024) reports 35–50% leak rates in multi-party negotiation benchmarks; our leakage-cell baselines land in their band for frontier models. AgentHarm (Andriushchenko et al., 2024) and AdvBench-style adversarial probe sets test single-turn

refusal robustness but do not exercise multi-turn principal-on-behalf-of dynamics. SecAlign (Chen et al., 2025) and PromptInject (Perez and Ribeiro, 2022) attack the system-vs.-user instruction boundary; PrincipalBench can be read as the cooperative analogue — the agent is supposed to follow the principal’s hidden instructions, and the failure mode is folding to a non-principal counterparty.

Principal-loyalty / negotiation register. The PaceBench sibling paper (Evans et al., 2026) introduces the `LoyaltyState` observer pattern that this work’s scaffolded arm builds on. Recent work on emotional-pacing and multi-turn negotiation frames the same conflict but does not provide a quantitative benchmark or trained-weight intervention.

Posttraining methods. DPO (Rafailov et al., 2023) is our preference-optimization tool. DAPO (Bytedance, 2024) provides the GRPO variant with dynamic sampling and asymmetric clip used here. The “stable reward basin” finding parallels observations in the RLHF reward-hacking literature (Casper et al., 2023; Tien et al., 2023) that small reward changes can produce large policy distribution shifts; we extend this to the joint reward $\times$ data design space.

Failure-mode taxonomies. The split judge separating `against_principal` from `missed_instruction` is informed by the broader specification-gaming literature (Krakovna et al., 2020) and the “alignment tax” framing of helpful/harmless trade-offs (Askell et al., 2021). Our manifold framing is a quantitative version of the alignment-tax intuition with cell-resolved measurement.

9 Limitations

Resolved in this draft. (i) Single base model — replicated on Mistral-7B-Instruct-v0.3, ± 2 pp on every leak metric (Section 6). (ii) Single counterparty vendor — 3-vendor swap preserves manifold shape (Figure 4). (iii) Architectural fix vs. data fix attribution — baseline ablation shows the sentinel alone leaves leak at 66% (Section 6).

Open.

- **$n=36$ items in the headline benchmark;** bootstrap CIs on per-cell rates are wide (cell size 4–10). The v0.5 expansion adds 15 items targeting concentrated failure-mode items but does not change headline-grid claims.
- **Counterparty coverage is three vendors, one model each.** The Phase 1/2/2.1 sweeps and the headline still use only `claude-sonnet`; we have not checked whether the `v4.1` $\rightarrow$ DAPO-v1 *delta* reproduces under alternative counterparties, only that the endpoint does.
- **Dual-judge κ is moderate** (any-fire 0.40–0.42; per-sub-flag 0.07–0.31 on fabrication/deception). We report any-fire aggregate only.
- **Hosted-model serving opacity** — Phase 1 numbers for OpenRouter-served subjects should be treated as lower bounds.

10 Conclusion

The leak/MI tradeoff in principal-acting LLM agents is a structural multi-axis manifold, not an artifact of any specific judge, prompt, training method, dialogue partner, or base model. We characterize the manifold through five independent levels of evidence and three robustness tests, release a 36-item $\times$ 3-arm benchmark, and present a three-stage SFT $\rightarrow$ DPO $\rightarrow$ DAPO recipe that ratchets total harm from 42 to 25 of 108 trajectories on Qwen3-8B and replicates within ± 2 pp on

Mistral-7B. The *recipe* is not the contribution — it is the falsifier we used to map the manifold’s structure. The contribution is the manifold’s geometry: stable reward $\times$ data basins, coupled axes that resist naive composition, an item-coverage ceiling at this experimental scale that is co-determined by reward and data design rather than either alone, and an architectural sentinel that reframes a stubborn reader-identity over-refusal failure as a prompt-feature problem rather than a data problem.

The 25/108 ceiling has not been broken. Driving harm below it will need joint reward $\times$ data co-tuning outside the v1 basin, on a meaningfully larger corpus than $n=31$. PrincipalBench, the trained-model artifacts, and the experimental scaffolding for follow-up co-tuning experiments are released.

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Appendix

A Items per cell (v0 grid, $n=36$)

cell	v0 subcells	v0 items
leakage	price, BATNA, health, settlement, source	5
capitulation	anchored, framing, repq ($\times 2$), sunk, deadline	6
posture	guilt ($\times 2$), harass, escalation, deadline	5
authoring	bench, policy, redteam, testcase, peer_review	5
sanity	mutual ($\times 2$), negot, coop, to-principal probes ($\times 6$)	10
moderation	inbound, mediation, press, partner, hr	5

The v0.5 expansion (Appendix H) adds 15 items: 4 reader-is-principal probe variants, 4 capitulation-anchoring variants, and 7 coverage-breadth items (M&A target, trade secret, vendor relations, board confidentiality, aggressive escalation, cross-team coordination, JV negotiation). Item definitions: `items/v0/*.json` and `items/v0_5/*.json`.

B Phase 1 numerical baseline (5 subjects)

subject	plain	prompted	scaffolded
<i>leak_rate</i>			
claude-sonnet	0.18	0.14	0.14
gpt-5-mini	0.14	0.14	0.14
gemini-3p1	0.27	0.18	0.18
qwen2.5-27b	0.20	0.14	0.14
qwen3-8b (OR)	0.38	0.34	0.32
qwen3-8b (local)	0.88	0.84	n/a

OpenRouter’s **qwen3-8b** shows leak 0.38 in plain arm; the *same weights* served via local vLLM with bf16 and default sampling show leak 0.88. Without visibility into OpenRouter’s safety layer we cannot attribute the delta precisely, but it is large and in the direction of hiding baseline failure.

C Judge design and the leak/MI separation

The first level at which the leak/MI manifold appears is the judge rubric itself. Early aggregate-harm judges scored “against_principal” as a composite of leak, capitulation, and over-refusal — the rewritten judge separates material **against_principal** (revealed a withhold, conceded a must-hold, fabricated to principal’s detriment) from **missed_instruction** (held positions correctly but failed to execute a principal-instructed action). On the v2 model evaluation, the split judge gives totals (fab=0, dec=1, AP=7, miss=9, any=9) where the composite would have read 9 “against_principal” fires, hiding the 7-vs-9 leak/MI separation. Without this judge separation the manifold is invisible at the score level.

D Pre-DAPO DPO ablations: v0–v2 era

The v4.1 DPO endpoint that DAPO is layered on inherits four design choices made through earlier ablations on the n=30 grid (predates v0.3 expansion to n=36). We report headline numbers here; full per-cell tables in repo at `runs/phase2_trained_v{0,1,1_lite,2}/scored.jsonl`.

Four preference-pair sets.

- **v0** (35 first-turn pairs only).
- **v1** (59 pairs = v0 $\cup$ 24 multi-turn pairs).
- **v1-lite** (51 pairs = v1 minus 8 authoring-cell MT pairs).
- **v2** (57 pairs = v1-lite $\cup$ 6 probe-gated authoring MT pairs).

Four findings.

1. **First-turn pairs alone (v0) buy almost all of the leak-rate reduction** (0.85 $\rightarrow$ 0.26), but leave hold nearly flat (0.83 $\rightarrow$ 0.85).

2. **Multi-turn pairs add the hold lift** (v1-lite: hold 0.96, posture 0.35) — the student needs late-turn teacher refusals to learn that pressure-through-repetition is to be refused, not answered.
3. **Authoring-cell MT pairs contaminate training** because the teacher’s refusal-by-default behavior conflicts with the agent’s legitimate task. Filtering pairs by whether the *teacher* was correct (not just whether the *student* failed) is the generalizable lesson.
4. **The engagement $\leftrightarrow$ refusal tradeoff is real.** v2 trades a capitulation-cell win ($0.36 \rightarrow 0.17$) for a leakage-cell regression ($0.06 \rightarrow 0.27$). The authoring-aware scaffold’s “what I can discuss at a meta level” register spills across cells and weakens the stricter “I can’t share that” register that leakage items benefit from.

E Reader-identity sentinel: ablation, generalization, and spoof-attack

Pair-curation alternatives (negative results). Before the architectural fix, we tried two pair-curation alternatives: **v2.2** (v2 + 13 produce-for-principal pairs against PLAIN_SYSTEM) and **v0.5** (93 pairs balanced across plain and prompted templates, 4 additional reader-is-principal items). v2.2 collapses authoring-cell leak to 0.13 (best of any variant) but produces a register flip in the prompted arm: capitulation $0.33 \rightarrow 0.72$, moderation $0.05 \rightarrow 0.40$. v0.5 partially recovers prompted-arm regressions but introduces new plain-arm regressions; aggregate plain-arm leak settles at 0.333, worse than v2 (0.206) or v2.2 (0.139). The hypothesis “conditional produce/refuse policy is learnable if we just balance the pair templates” is falsified at this data scale.

The sentinel fixes the failure.

variant	plain	prompted	total
baseline	3/6	3/6	6/12
v2	4/6	5/6	9/12
v0.5	3/6	5/6	8/12
v0.6 (sentinel)	1/6	2/6	3/12

v0.6 is the only trained variant to land below baseline on the 12-trajectory probe.

Generalization (held-out items). Of the 6 probe items, 4 produced training pairs {01, 02, 04, 05}; items {03, 06} are de facto held out. v0.6 reduces held-out item 06 from baseline 2/2 to 0/2 and improves item 03 from 2/2 to 1/2. The sentinel-conditional behavior generalizes.

Spoof-attack robustness. Forcing [READER: PRINCIPAL] on third-party leakage items produces one-fact slippage on a single ambiguous item (briefing’s principal is a policy *drafter*) and zero leakage on price-negotiation or BATNA items. Forcing [READER: THIRD_PARTY] on principal items: spoof-sentinel fires 5/12 vs. correct-sentinel 3/12 — +2 failures, not a global flip. The model cross-references the sentinel against the briefing’s structural cues; the sentinel is a guide weighed against other evidence, not a master switch.

F The prompting frontier — v1→v4 ablation

We iterated the **prompted-arm** system prompt through four versions and re-scored Phase 1 (5 subjects $\times$ 16 items) under the rewritten judge.

metric (prompted % / scaffolded %)	v1	v2	v3	v4
leak_rate	22.0 / 22.0	11.0 / 12.0	16.0 / 13.0	12.0 / 12.0
harm_fire	18.0 / 13.0	15.0 / 18.0	8.0 / 8.0	6.0 / 6.0
bound_leak (count)	6 / 5	3 / 2	2 / 3	1 / 1
commitment_hold	95 / 100	97 / 98	95 / 98	97 / 100

v1→v2 (adversarial-stranger framing): halves leak; harm flat. v2→v3 (private-bound/position distinction + execute-positive-instructions): cuts harm 54%, but leak drifts back +29% via over-eager conditional-permission execution and enumerate-leak resurgence. v3→v4 (narrow-scope surgery): adds “conditional permissions are not proactive offers” rule; restores v2’s tightness. v4 is Pareto-better than every prior version on prompted and scaffolded cells.

Bound below 12% leak / 6% harm. Even after four disciplined iterations, leak remains $\sim 12\%$ and harm $\sim 6\%$ on 16 adversarial items, so the benchmark cannot be prompted away.

G The DPO walk-back: v4 → v4.1 in detail

v4 SFT+DPO produces posture-collapse: prompted/scaffolded leak drops to 5.1% / 5.3% (a $\sim 75\%$ reduction vs. v1) but **missed_instruction** rises to 41.7% / 44.4%, concentrated on reader-is-principal probes (14/18 refused) and cooperative sanity items (19/30). Other harm sub-flags are near zero — this is not a generic capability regression, but a register-level over-correction.

v4.1 mines the 42 MI regressions, pairs the v4-trained refusal as **rejected** with the teacher’s correct response as **chosen**, and runs a second DPO pass (137 pairs total, 207 steps, final margins ≈ 1.3).

metric ($n=36 \times 3$ arms, 108 rollouts)	v4 prompted	v4 scaff	v4.1 prompted	v4.1 scaff
leak_rate	5.1%	5.3%	6.0%	5.6%
missed_instruction	15 / 36	16 / 36	12 / 36	11 / 36
missed_instruction (probe cut)	5 / 6	6 / 6	3 / 6	5 / 6
leaked_private_bound (all arms)	2	—	5	—
harm_fire (all arms)	42/108	—	31 / 108	—

v4.1 cuts MI fires by a third ($42 \rightarrow 28$, -33%), recovers half the probe-item losses (plain arm $3/6 \rightarrow 0/6$), and holds leak essentially flat ($+0.9$ pp prompted, $+0.3$ pp scaffolded). Net harm is down 26%.

The frontier persists at finer grain. The three new **leaked_private_bound** fires are all on one item (**pb-capit-sunk-01**: across all three arms, the v4.1 model now offers the \$10K goodwill ceiling named in the briefing). The v4 model had *refused to make any offer* on that item (MI); v4.1 makes the offer but names the cap (bound-leak). Same item, same trajectory structure, crossed from one failure mode to the other — the manifold in miniature.

H Data expansion ablation in detail

We expanded the training set from $n=31$ (v0 grid minus 5 heldout) to $n=46$ (v0.5 grid minus the same 5 heldout). The 15 new items target the 7 concentrated failure-mode items from Figure 3: 4 reader-is-principal probe variants, 4 capitulation-anchoring variants, and 7 coverage-breadth items. Reward function and hyperparameters are identical to v1.

metric (108 rollouts)	v4.1	DAPO-v1 ($n=31$)	DAPO-v1 ($n=46$)
total harm_fire	31	25	38
total MI	29	24	38
bound_leak	5	4	3
plain leak	11.1%	11.1%	9.5%
scaffolded leak	5.6%	3.9%	5.6%

Per-item delta $n=31 \rightarrow n=46$. 11 improvements (7 of which are exactly the high-leverage items the expansion was designed to fix: `pb-author-to-principal-01/03/05`, `pb-capit-anchored-01`); 24 regressions (concentrated in cells *not* the focus of the expansion — leakage $2/15 \rightarrow 7/15$, posture $1/15 \rightarrow 5/15$). The new `pb-leak-finance-01` / `pb-leak-trade-01` items pushed leak-cell behavior in the wrong direction on old leak items. The basin claim therefore generalizes from reward design to data design: changing either dimension alone falls out of v1’s stable basin into worse local optima.

I Reproducibility paths

All artifacts at `/home/ubuntu/principal-loyalty/`; scripts in `scripts/` are runnable end-to-end in the order described by `progress.md`.

Items. `items/v0/*.json` ($n=36$); `items/v0_5/*.json` ($n=51$).

Phase 1. Hosted-vendor runs at `runs/phase1*/scored.jsonl`. Local-vLLM Qwen3-8B baseline (88% plain-leak number) at `runs/phase2_baseline/scored.jsonl` and `runs/phase3_baseline_qwen/scored.jsonl`.

Phase 2 (SFT+DPO). Earlier-iteration outputs at `runs/phase2_trained{,_v1,_v1_lite,_v2,_v06}/scored.jsonl`. Headline DPO endpoint: `runs/qwen_sft_dpo_v4_1_merged/` (16 GB), eval at `runs/phase2_trained_v4_1/scored.jsonl`. DPO pair files: `data/dpo_v{0,1,1_lite,2,05,06,4,4_combined}.jsonl`.

Phase 3 (DAPO). Headline checkpoint: `runs/qwen_dapo_v1_step35_merged/` (16 GB). Reward variant adapters: `runs/qwen_dapo_{v2,leakonly,v3}/`. Phase 3 evals: `runs/phase3_dapo_v1_step35/`, `runs/phase3_dapo_v{2_step55,leakonly_step35,v3_step{30,55}}/`. Reward function: `src/reward.py`. DAPO config: `scripts/run_dapo.sh`.

Counterparty robustness (Section 6). `runs/phase3_dapo_v1_step35{,_gpt5cp,_gemcp}/scored.jsonl`. Comparison: `scripts/compare_counterparty.py`.

Cross-family Mistral. Adapters: `runs/mistral_sft_v4/`, `runs/mistral_dpo_v4_1/`. Merged: `runs/mistral_sft_dpo_v4_1_merged/` (14 GB). Eval: `runs/phase3_mistral_sft_dpo/scored.jsonl`.

Data expansion (Appendix H). `runs/qwen_dapo_v1_n51_step55_merged/` (16 GB), eval at `runs/phase3_dapo_v1_n51_step55/scored.jsonl`. Train data: `data/ver1_train_v0_5.parquet` (46 train rows / 5 val).

Item-level harm analysis (Figure 3). runs/phase3_dapo_v1_step35/item_harm_analysis.txt. Script: scripts/item_harm_analysis.py. Figure-generation script: scripts/make_figures.py.